Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/03/2024
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rohit.k.trivedi@gmail.com	**Fax:**	FY 515-294-4291
PI Organization Type:	UNIVERSITY	**Phone:**	515-450-1948
Organization Name:	Iowa State University		
PI Address 1:	Dept. of Materials Science and Engineering		
PI Address 2:	100 Wilhelm Hall		
PI Web Page:			
City:	Ames	**State:**	IA
Zip Code:	50011	**Congressional District:**	4
Comments:	Other email: trivedi@ameslab.gov		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/22/2016	**End Date:**	06/21/2021
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
Contact Email:	louise.s@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/21/2021 per NSSC information (Ed., 8/7/19)		
Key Personnel Changes/Previous PI:	January 2018 report: Prof. Alain Karma is no longer a Co-Investigator on this project, but continues to be a collaborator, who is responsible for the modeling part of the project through a separate NASA grant.		
COI Name (Institution):			
Grant/Contract No.:	NNX16AL77G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)" grant number NNX12AK54G, with the same Principal Investigator Dr. Rohit Trivedi. During the manufacturing of industrial components by solidification processes, which include casting, welding, Soldering and single crystal growth, the properties and the reliability of the finished products depend on the resulting microstructure that is governed by the processing conditions imposed during the solidification process. It is thus crucial to quantitatively relate the detailed microstructural features with the growth conditions and the composition of the alloy so that one can design alloy composition and growth conditions to optimize processing routes that will produce the object with superior properties. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Extensive experimental and theoretical studies over the past several decades have yielded basic insights into the formation of complex solidification patterns; their dynamics in three dimensions (3D) still remain poorly understood since fluid convection effects on Earth lead to large scale variation in temperature and alloy concentration in bulk samples that are quite complex to quantify. The presence of these convection effects make it difficult to probe the spatially extended pattern dynamics under well-controlled diffusive growth conditions with imposed constant growth velocity, thermal profile, and alloy composition. To minimize convection effects, detailed experimental studies of interface dynamics on Earth have been restricted to thin samples where convection effects are minimized. However, in thin samples it is not possible to study the formation of extended three-dimensional patterns. Also, the dynamics of microstructure evolution is strongly influenced by the thickness of the sample. Therefore, directional solidification experiments are designed on International Space Station (ISS) where convection effects are negligible so that precise evolution of three-dimensional microstructure can be investigated and characterized quantitatively. These results can then be used to develop appropriate theoretical models and to validate these models quantitatively to confirm the predictive nature of the models. Under conditions of very low velocity and low composition, with high thermal gradient, solidification occurs with a planar interface. The most common solid-liquid interface patterns that form during the solidification in metals, however, consist of complex spatially extended arrays of cells or dendrites that form when the growth conditions exceed the critical condition for the onset of morphological instability of a planar interface. In previous microgravity experiments, Directional Solidification Insert (DSI) experiments, we examined the time evolution of the three-dimensional patterns under a wide range of velocity for two different thermal gradients in succinonitrile-0.24 wt% camphor. For this dilute composition, the microstructures were mainly planar and cellular. Dendritic arrays were observed at higher velocities, although at higher velocities significant latent heat effects caused the macroscopic interface to become curved so that only very limited velocities could be examined in the dendritic regime. In the current study, DSI-R experiments, carried out on the International Space Station, a higher alloy composition, succinonitrile-0.5 wt % camphor, was used to examine the dynamics of 3D dendritic arrays at lower velocities. In addition, the effect of composition on the spatiotemporal evolution of three-dimensional cellular and dendritic array structures will also be characterized. Since one of the goals is to examine the formation of dendritic arrays that are characterized by the presence of side branches, specific emphasis was placed on the fundamental understanding of the origin of side branch formation and the subsequent dynamics of side branch propagation in the two-phase region or the mushy zone, between the primary dendrite arms. These sidebranch patterns are directly related to the scale of microsegregation that controls the properties and reliability of cast objects. Detailed experimental studies were carried out in a microgravity environment where convection is reduced sufficiently to obtain growth under diffusive conditions. The interface evolution in three dimensions was directly visualized through two cameras that viewed the interface from the side and from the top. This continuous observation of the interface provided an unprecedented opportunity to elucidate fundamental mechanisms of the formation of those hierarchical structures by providing quantitative benchmark data that can be compared to predictions of computational models. The goal of this work is therefore to (i) obtain extensive benchmark experimental data under microgravity conditions over a range of velocity, thermal gradient, and alloy composition in a transparent system by imaging the dynamic evolution of microstructures under directional solidification conditions in the Directional Solidification Insert (DSI) of the DECLIC facility (DEvice for the study of Critical LIquids and Crystallization) installed onboard the ISS, and (ii) use state-of-the-art phase-field simulations to make quantitative predictions of dendritic hierarchical structure formation under diffusive growth conditions that can be directly compared to experimental observations under prescribed growth conditions. We expect the combination of those experimental and computational studies to establish a reliable framework for predicting dendritic microstructures in a wide range of alloy systems of technological relevance. This research project is carried out in collaboration between NASA and CNES (Centre national d'études spatiales (CNES), the French government space agency) and includes experiments on International Station, experiments under terrestrial conditions in thin and bulk samples, and the development of rigorous theoretical models to compare with the experimental data obtained terrestrially in thin samples and from International Space Station.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based super alloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process that is used to fabricate appropriate components. These microstructures in turn control the integrity and mechanical properties of cast parts. By combining well-controlled and accurately characterized experimental evolution of three-dimensional solidification microstructures with quantitative phase-field simulations, this research should bring significant progress in understanding the origin and propagation of side-branching and the nature of the cell to dendrite transition, which have been a long standing issue in solidification for several decades. The deeper understanding of microstructure formation dynamics, including the development of side-branched structure in the mushy zone, shall thus accelerate and optimize the design of new technological alloys and their processing conditions that should directly benefit US industries.

Task Progress:	Two different processing routes were used to obtain coherent side branches in directionally solidified alloy. Experiments were carried out to characterize side branches as a function of increasing thermal gradient. At low thermal gradient, incoherent side branches form due to the amplification of thermal noise at the tip. However, as the thermal gradient is increased, the first side branch forms closer to the tip and at some critical gradient the solute field of the side branch influences the tip shape or tip radius. We characterized critical thermal gradient values for three different velocities. The critical thermal gradient was found to decrease as the velocity was increased. Experiments were designed with oscillatory imposed velocity. Initially the alloy was solidified at a constant velocity and then oscillatory velocities with different frequencies were imposed. A critical frequency at each velocity was determined when coherent side branches initiated. The microstructure features showed that when coherent side branches formed, the first side branch location did not change so that the mechanism of side branch formation is different from that of the gradient effect in which the first side branch forms closer to the tip when side branches form closer to the tip.
Bibliography Type:	Description: (Last Updated: 04/03/2024)
Abstracts for Journals and Proceedings	Ji K, Mota FL, Strutzenberg LL, Trivedi R, Bergeon N, Karma A. "Characterization of dendritic spatially extended 3D patterns in directional solidification: Microgravity experiments in DECLIC-DSI onboard ISS and 3D phase-field simulations." TMS 2021. 150th Annual Meeting, The Minerals, Metals and Materials Society, Orlando, Florida, March 15-18, 2021. Abstracts. TMS 2021. 150th Annual Meeting, The Minerals, Metals and Materials Society, Orlando, Florida, March 15-18, 2021. , Mar-2021
Abstracts for Journals and Proceedings	Ji K, Mota FL, Song Y, Lyons T, Strutzenberg LL, Trivedi R, Bergeon N, Karma A. "Three-dimensional in situ observations of polycrystalline microstructure evolution during directional solidification of transparent alloys aboard the ISS and quantitative comparison with numerical modeling." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
NASA Technical Documents	Ji K, Mota FL, Strutzenberg LL, Trivedi R, Bergeon N, Karma A. "Characterization of dendritic spatially extended 3D patterns in directional solidification: Microgravity experiments in DECLIC-DSI onboard ISS and 3D phase-field simulations." Huntsville, Alabama: Marshall Space Flight Center, 2021. NASA/Document ID: 20210010710. https://ntrs.nasa.gov/citations/20210010710 , Feb-2021
Papers from Meeting Proceedings	Mota FL, Medjkoune M, Ji K, Littles LS, Trivedi R, Karma A, Bergeon N. "Dynamics of solidification microstructure formation in DECLIC-DSI onboard ISS: dendritic patterns data treatment." IAC 2022: 73rd International Astronautical Congress, Paris, September 18-22, 2022. Abstracts. IAC 2022: 73rd International Astronautical Congress, Paris, September 18-22, 2022. IAC Paper ID 67562. https://dl.iafastro.directory/event/IAC-2022/paper/67562/ , Sep-2022